VSCPU / CARAVEL

Custom Development Board • Pinout Diagram

J3 / APPLICATION GPIO

		QFN pin
+3.3V	3.3 V output	—
GPIO_IN_1	IO16 · input bit 0 (W8255[0])	53
GPIO_IN_0	IO17 · input bit 1 (W8255[1])	54
GPIO_OUT_0	IO30 · output bit 3 (W8254[3])	7
GPIO_OUT_1	IO29 · output bit 2 (W8254[2])	6
GPIO_OUT_2	IO28 · output bit 1 (W8254[1])	5
GPIO_OUT_3	IO27 · output bit 0 (W8254[0])	4
GND	Ground	—

USING THE GPIO PINS

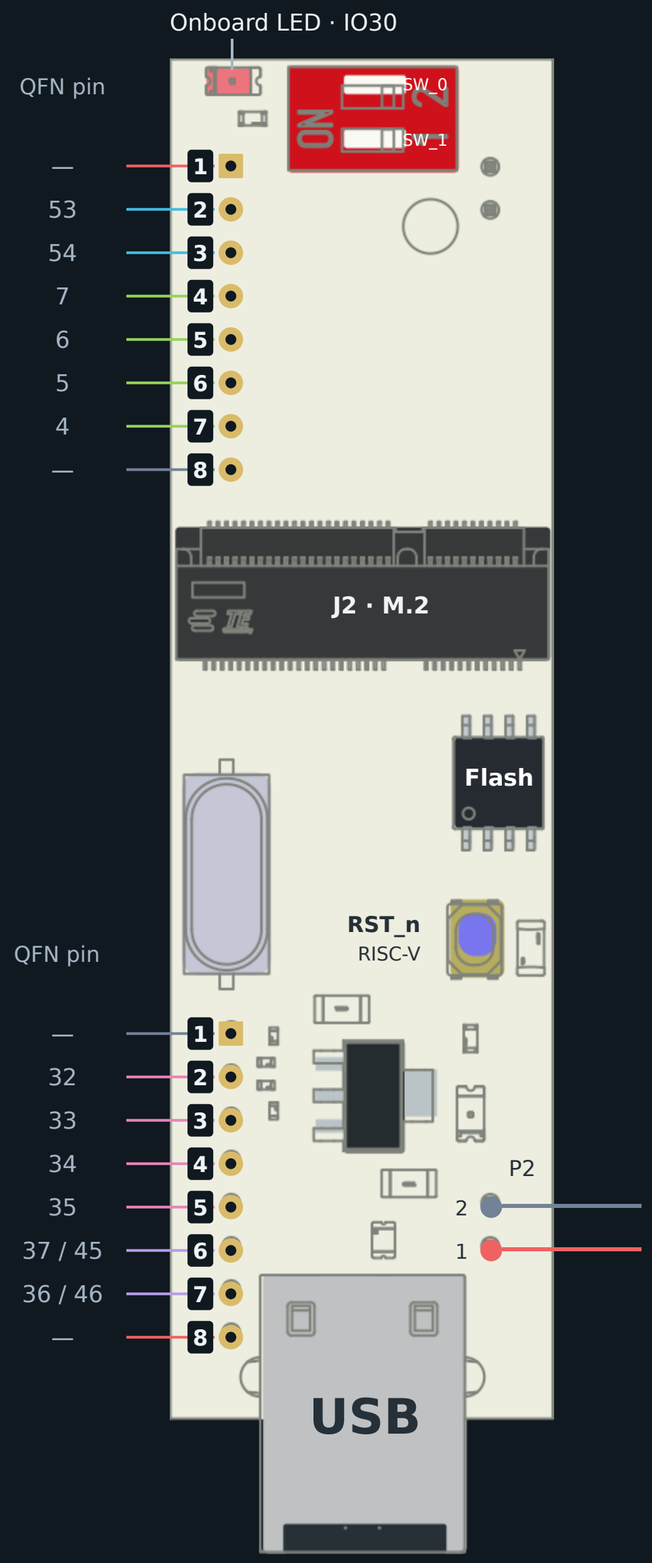
W8254[3] = bit 3 of word address 8254 (decimal).
This is register notation, not ASM or C syntax.

`CPi 8254 8`	J3.4 HIGH; other GPIO outputs LOW.
`CP 48 8255`	Read inputs into RAM word 48.
`ANDi 48 1`	Keep bit 0: J3.2 reads as 0 or 1.

3.3 V logic · common GND · input mask: 1 << bit number.
Example: vscpu3x_apps/gpio_test/ (C + ASM)

J1 / PROGRAMMING + UART

		QFN pin
GND	Ground	—
SDO	IO1 · Caravel → host (MISO)	32
SDI	IO2 · Host → Caravel (MOSI)	33
CS_n	IO3 · Active-low SPI select	34
SCK (CLK)	IO4 · Housekeeping SPI clock	35
TTL_TX	IO6 (management TX) ↔ IO11 (VSCPU RX)	37 / 45
TTL_RX	IO5 (management RX) ↔ IO12 (VSCPU TX)	36 / 46
+3.3V	3.3 V rail	—



MODE + RESET

SW2: two-bit selector

Bit order: SW_1 SW_0

00 Run
01 Program codemaker (CM)
10 Program control tower (CT)
11 Program agent 0 (A0)

VSCPU RESET

Select 01 → wait for acceptance → select 00.
Firmware pulses IO24 → IO8, then releases run mode.

VSCPU reset
IO24 → IO8



Programming mode gates the cores during the pulse.
Timing is illustrative; software delay varies with clock.

RST_n button: active-low RISC-V reset (QFN 21).

Onboard regulators: 5 V → 3.3 V → 1.8 V.

GND · P2.2
+5 V · P2.1
Direct rail input; bypasses the USB fuse.

Power · Ground · GPIO input · GPIO output · SPI · UART · Control

Orientation: top view.